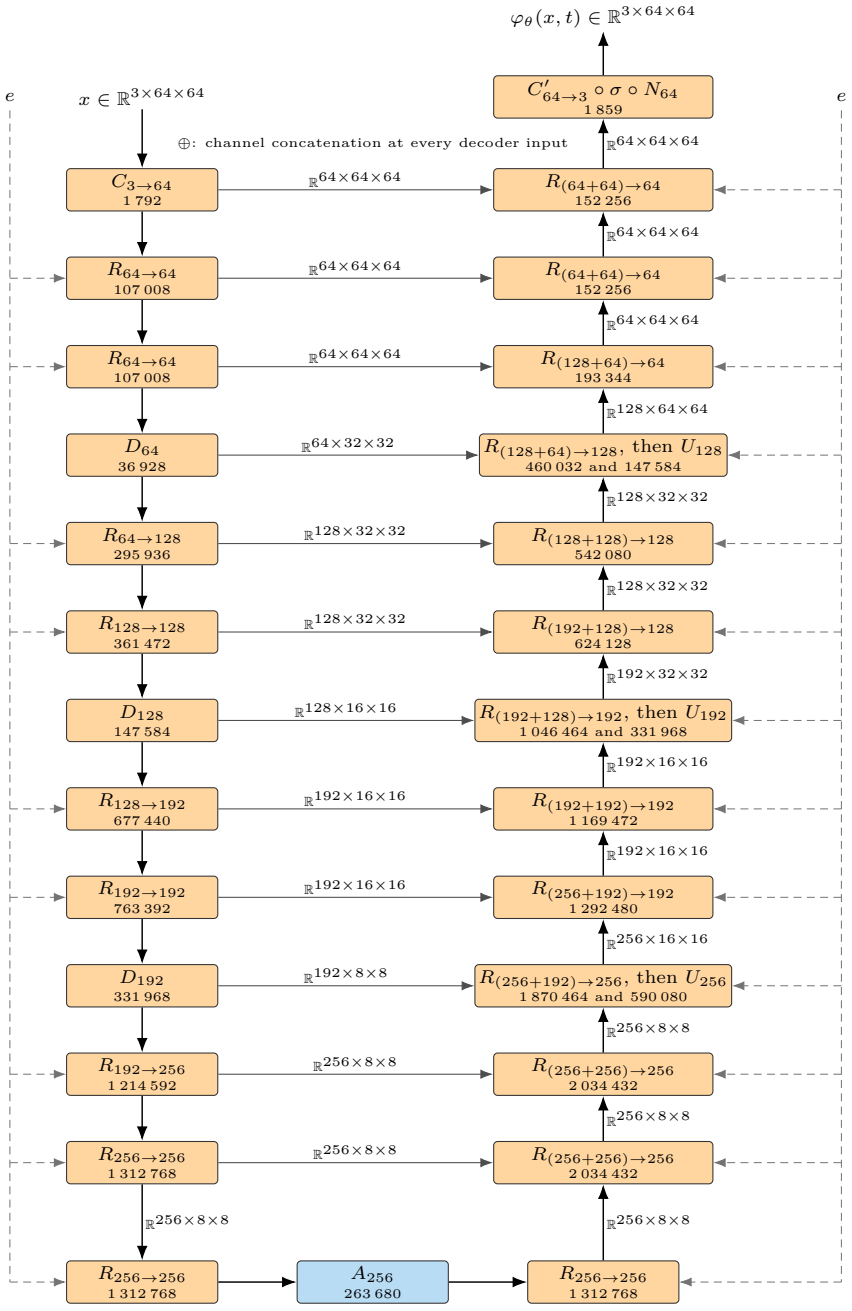
The network $\varphi_\theta : \mathbb{R}^{3\times64\times64} \times \mathbb{R} \to \mathbb{R}^{3\times64\times64}$ used for CelebA at 64×64 (channels $\times$ height $\times$ width), 20 972 867 parameters. Each box shows its symbol and parameter count; the space on an arrow is the codomain of the map the arrow leaves. Dashed arrows carry $e = E(t) \in \mathbb{R}^{256}$ into every residual block.



TIME CONDITIONING. $t \in \mathbb{R} \xrightarrow{\tau} \mathbb{R}^{64} \xrightarrow{L_{64\to256}} \mathbb{R}^{256} \xrightarrow{\sigma} \mathbb{R}^{256} \xrightarrow{L_{256\to256}} e = E(t) \in \mathbb{R}^{256}$

$\tau(t) = (\cos\omega_i t)_{i=0}^{31} \oplus (\sin\omega_i t)_{i=0}^{31} \in \mathbb{R}^{64}$, $\omega_i = 10^{-4i/32}$; E has 82 432 parameters. τ is the sinusoidal positional table of a transformer evaluated at the real argument t instead of an integer position (same frequencies, same cosine and sine halves).

RESIDUAL BLOCK $R_{a\to b} : \mathbb{R}^{a\times h\times w} \times \mathbb{R}^{256} \to \mathbb{R}^{b\times h\times w}$:

$$R_{a\to b}(z,e) = S(z) + C'_{b\to b}\Big(\sigma\big(N_b\big(C_{a\to b}(\sigma(N_a(z)))\big) \odot (1+s(e)) + u(e)\big)\Big), \qquad [s(e);u(e)] = L_{256\to2b}(\sigma(e)) \in \mathbb{R}^{2b},$$

with $S = \mathrm{id}$ if $a = b$ and $S = C^1_{a\to b}$ otherwise; $s(e), u(e) \in \mathbb{R}^b$ scale and shift channels, broadcast over pixels; C' is zero at initialisation (dropout rate 0). Parameters: $2a + 9ab + 9b^2 + 518b + [a \neq b]\,(ab + b)$.

ATTENTION BLOCK $A_a : \mathbb{R}^{a\times h\times w} \to \mathbb{R}^{a\times h\times w}$, used once with $a = 256$ on $m = hw = 64$ tokens: z is reshaped to $X \in \mathbb{R}^{a\times m}$ and

$$[Q;K;V] = W_{qkv}\,N_a(X) + b_{qkv}, \qquad H_j = V_j\ \mathrm{softmax}\Big(\frac{K_j^\top Q_j}{\sqrt{a/4}}\Big), \qquad A_a(X) = X + W_o\,[H_1;\ldots;H_4] + b_o,$$

where $Q_j, K_j, V_j \in \mathbb{R}^{a/4\times m}$ are the rows of head $j = 1,\ldots,4$, the softmax runs over the key index, $W_{qkv} \in \mathbb{R}^{3a\times a}$, $W_o \in \mathbb{R}^{a\times a}$ (zero at initialisation). Parameters: $4a^2 + 6a$.

COMPONENTS.

symbol	definition	domain $\to$ codomain	parameters
τ	sinusoidal embedding (above)	$\mathbb{R} \to \mathbb{R}^{64}$	0
$L_{a\to b}$	$x \mapsto Wx + b$	$\mathbb{R}^a \to \mathbb{R}^b$; $(a,b) \in \{(64,256),(256,256)\}$ in E and $(256,2b)$ inside every $R_{\cdot\to b}$	$ab + b$
σ	$x \mapsto x/(1+e^{-x})$ entrywise	$\mathbb{R}^n \to \mathbb{R}^n$	0
N_a	group normalisation: $(z-\mu_g)/\sqrt{v_g+\varepsilon}$ over 32 groups g of $a/32$ channels (statistics over the group's channels and all pixels), then per-channel $\gamma_c z + \beta_c$	$\mathbb{R}^{a\times h\times w} \to \mathbb{R}^{a\times h\times w}$	$2a$
$C_{a\to b}$	3×3 convolution, stride 1, unit zero padding (C': zero at initialisation)	$\mathbb{R}^{a\times h\times w} \to \mathbb{R}^{b\times h\times w}$	$9ab + b$
$C^1_{a\to b}$	1×1 convolution, i.e. $L_{a\to b}$ applied at every pixel	$\mathbb{R}^{a\times h\times w} \to \mathbb{R}^{b\times h\times w}$	$ab + b$
D_a	3×3 convolution, stride 2, unit zero padding	$\mathbb{R}^{a\times h\times w} \to \mathbb{R}^{a\times h/2\times w/2}$	$9a^2 + a$
U_a	nearest-neighbour upsampling by 2 (parameter-free), then $C_{a\to a}$	$\mathbb{R}^{a\times h\times w} \to \mathbb{R}^{a\times 2h\times 2w}$	$9a^2 + a$
$\oplus$	channel concatenation (every decoder input)	$\mathbb{R}^{a\times h\times w} \times \mathbb{R}^{b\times h\times w} \to \mathbb{R}^{(a+b)\times h\times w}$	0
$R_{a\to b}$	residual block (formula above)	$\mathbb{R}^{a\times h\times w} \times \mathbb{R}^{256} \to \mathbb{R}^{b\times h\times w}$	see above
A_a	attention block (formula above)	$\mathbb{R}^{a\times h\times w} \to \mathbb{R}^{a\times h\times w}$	$4a^2 + 6a$
head	$C'_{64\to3} \circ \sigma \circ N_{64}$	$\mathbb{R}^{64\times64\times64} \to \mathbb{R}^{3\times64\times64}$	1 859

COLOUR LEGEND. ■ exists as an object of the sparse-transformers classes: an affine map applied per token or per pixel is a sparse linear layer with fan-in K ($K = n_{\text{in}}$ is the dense layer): the time path, the scale-shift maps, the 1×1 convolutions, the attention projections. ■ could reasonably be made sparse but is not one of those objects: a 3×3 convolution is a parameter-free unfold (gather of the $3\times3\times a$ neighbourhood of a pixel) followed by a per-pixel linear map $\mathbb{R}^{9a} \to \mathbb{R}^b$, a sparse linear layer with $K \le 9a$, but with weights shared across pixels and a structured, not random, support. ■ could be substituted: softmax attention by attention-free attention or windowed causal attention (both causal), group normalisation by dynamic-tanh or root-mean-square normalisation. ■ parameter-free. A composite box (R, D, U, head) takes the colour of its weight-carrying parts; its formula colours each factor.

FAN-IN. A sparse linear layer $S : \mathbb{R}^{n_{\text{in}}} \to \mathbb{R}^{n_{\text{out}}}$ with fan-in K computes $S(x) = (M \odot W)\,x + b$ with $M \in \{0,1\}^{n_{\text{out}}\times n_{\text{in}}}$, $\sum_j M_{ij} = K$ for every i, the K nonzero entries of each row drawn uniformly without replacement, $W \in \mathbb{R}^{n_{\text{out}}\times n_{\text{in}}}$ trained and $\odot$ entrywise; K is clamped to $1 \le K \le n_{\text{in}}$ and $K = n_{\text{in}}$ gives the dense map L (the masked dense weight and the $n_{\text{out}} \times K$ weight with an index table are two implementations of the same map). In the graph whose edges are the pairs (i,j) with $M_{ij} = 1$ the constraint fixes in-degrees only:

$$k_i^- = \sum_j M_{ij} = K \text{ (output unit } i\text{)}, \qquad d_j^+ = \sum_i M_{ij} \text{ (input unit } j\text{)}, \qquad \mathbb{E}\,d_j^+ = \frac{n_{\text{out}}K}{n_{\text{in}}}.$$

In a stack of layers $l = 1,\ldots,L$ with widths $n_0,\ldots,n_L$ and fan-ins K_l, a hidden unit v of layer l ($0 < l < L$) has degree $k_v = K_l + d_v^+$, so the maximum degree of the network graph obeys

$$K_G := \max_v k_v \ \ge\ \max_l K_l, \qquad \mathbb{E}\,k_v = K_l + \frac{n_{l+1}K_{l+1}}{n_l},$$

with equality only in the degenerate case where every hidden unit of layer l has out-degree at most $\max_{l'} K_{l'} - K_l$, which the random supports do not enforce. The fan-in K (per layer, K_l) is therefore the quantity of interest, not K_G; a green box is a map of the form S with any $K \le n_{\text{in}}$.